CODEZEN

Basic Details

Problem Statement ID: **10**

Problem Statement Title: **Twenty years of skill and no way to prove it**

Theme: **Skilled Trades, Livelihoods & Trust**

PS Category: **Software**

Team ID: **2392608084**

Team Name: **CODEZEN**

Team Members

Roll No.	Name
2392608084	Samriddhi Rathore
2392608072	Soumya Sarkar
2392608227	Maanya Ashchary
2392608182	Princy Suman
2392608278	Shagun Sagar
2392608092	Isha Bar

Shramik: Challenges & Proposed Solution

Challenges & Problems

- **No portable proof** — decades of skill, nothing that travels with the worker.
- **Restart from zero** — a new city means no references and no work record.
- **Word-of-mouth hiring** — contractors rely on informal networks.
- **Uncertain reliability** — no verified record of attendance or repeat hiring.
- **Fake claims** — self-written ratings are easy to manipulate.
- **Digital divide** — basic phones, low literacy, limited internet.

Proposed Solution

- **Verified Job Ledger** — worker logs a finished job by SMS, USSD or WhatsApp; the hirer confirms with a one-time code. Only confirmed jobs count.
- **Trust Score & Shramik Card:** built from completed jobs, repeat hirers ("asked back") and showing up on time. Shared as a QR / short link / PDF.
- **Hirer View:** a stranger scans the QR and sees a verified work history in seconds, with no app or login needed.

Innovation & Uniqueness

Worker completes job → "Asked back" signal (Repeat hires weigh most) → Job added to tamper-evident ledger → Trust score updates (completed jobs + repeat hires + punctuality) → QR Shramik Card generated/shared → New hirer scans QR and checks verified history

Basic-phone first: SMS / USSD / voice • local languages

Contractor vouching: trusted peers endorse skills.

Feasibility and Viability

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Skill • Work • Trust

Shramik: Feasibility & Viability

Technical Feasibility



- SMS/USSD/WhatsApp APIs are mature and cheap.
- Hash-chained ledger without heavy blockchain.
- Simple, explainable ML scoring.

Financial Feasibility



- Low running cost: SMS + small cloud server.
- Pilot grants, CSR and skilling schemes.
- Revenue from hirer verification and platform partners.

Operational Feasibility



- Roll out through contractors, hardware shops and labour chowks.
- Hirer confirmation is a 10-second step.
- Local-language voice prompts.

Social Viability



- Worker dignity and bargaining power.
- More trust and fewer no-shows.
- Workers own and carry their record.

Strategies for Overcoming Challenges

Fake Confirmations



- Hirer-phone verification, anomaly detection, unique-hire weighting.

Low Connectivity / Literacy



- Offline-friendly SMS/USSD, voice prompts, icon-based PWA and field agents.

Cold Start & Adoption



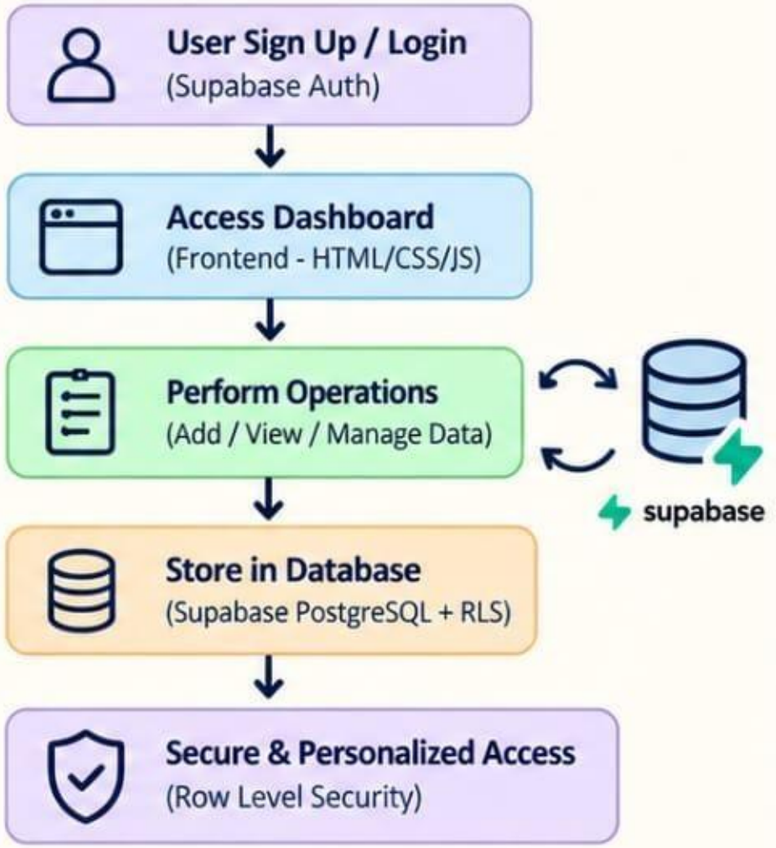
- Start in one city and 2–3 trades; onboard contractors first, then workers.

Technical Approach

Methodology and Process

1. **Plan**
   Understand requirements, define features and system design.
2. **Develop**
   Build frontend and backend using the chosen tech stack.
3. **Test**
   Validate functionality, security and performance.
4. **Deploy**
   Host on cloud and configure database and live environment.
5. **Maintain**
   Monitor, gather feedback and improve.

System Flow



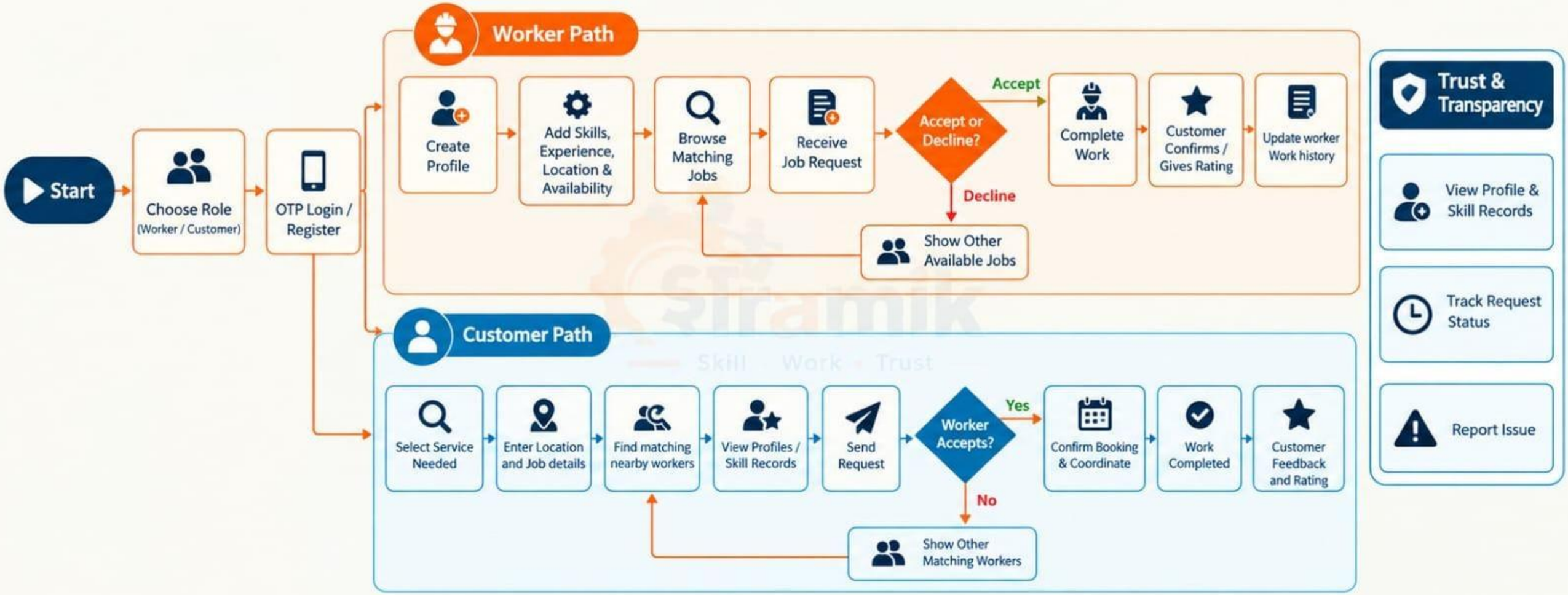
Technology Stack

- **Python** — Backend Logic
- **FastAPI** — API Framework
- **PostgreSQL** — Database
- **React** — Frontend
- **scikit-learn** — ML Models
- **SMS/IVR** — Notifications
- **Docker** — Deployment
- **Cloud** — (AWS / Azure)

Platform Flow

Connecting skilled workers with people who need reliable services.

Shramik — Skill • Work • Trust



Impact and Benefits

Workers

- Portable, provable record of skill.
- Fair access to new work in any city.

Hirers & Contractors

- Hire strangers with confidence.
- Fewer no-shows and disputes.

Society & Economy

- Formalises informal labour.
- Supports financial inclusion and credit access.

Government & NGOs

- Verified data for skilling schemes and e-Shram linkage.
- Better workforce insights.

Business Model

Target Customers

- Workers: painters, carpenters, tailors, electricians.
- Contractors, builders, housing societies and households.
- Skilling bodies, NGOs, municipalities.

Revenue and Funding

- Free for workers, always.
- Pay-per-verification for hirers and businesses.
- Grants, CSR and government skilling partnerships.

Scalability

- Pilot: one city, 2–3 trades.
- Expand to more trades and cities.
- Open API for job platforms, lenders and schemes.

Research and References

Reference Platforms & Standards

- **e-Shram** — national unorganised-worker registry
- **Urban Company** — verified service marketplace
- **W3C Verifiable Credentials** — portable tamper-evident proofs
- **DigiLocker** — portable digital documents

Key Insights

- Trust grows through repeat work, not self-claims.
- Phone-number identity and SMS reach most workers.
- Two-sided confirmation makes fake records costly.

Live Prototype



OPEN PROTOTYPE →

Explore the Shramik demo

https://ai.studio/apps/e75b1c0-dd3-4cfd-9a82-fddcbc35157a

Research & References

- **e-Shram** — eshram.gov.in
- **Urban Company** — urbancompany.com
- **W3C Verifiable Credentials** — w3.org/TR/vc-data-model-2.0/
- **DigiLocker** — digilocker.gov.in
- **Prototype (Shramik)** — ai.studio/apps/e75b1c0-dd3-4cfd-9a82-fddcbc35157a